PROJECT-Q Workbook - 12

30-Day Quantum Computing Challenge

DAY 12

Quantum Teleportation

Learning Objectives

By the end of this chapter, you will be able to:

- Explain what Quantum Teleportation actually means.
- Differentiate between teleporting a particle and teleporting a quantum state.
- Understand why entanglement is essential for teleportation.
- Describe the complete quantum teleportation protocol step by step.
- Explain the role of classical communication in the protocol.
- Interpret the quantum circuit used for teleportation.
- Implement Quantum Teleportation using Qiskit.

Quick Recap

Before learning Quantum Teleportation, let's connect the concepts you've already mastered.

- **Superposition** allows a qubit to exist in multiple states simultaneously.
- **Quantum Gates** manipulate quantum states.
- **Multi-Qubit Systems** allow qubits to interact.
- **CNOT Gate** creates correlations between qubits.
- **Entanglement** creates a strong quantum connection that cannot be explained classically.

Today's topic combines **all of these ideas** into one complete communication protocol.

Superposition
↓
Quantum Gates
↓
Multi-Qubit Systems

↓
Entanglement
↓
Quantum Teleportation

Why This Topic Matters

Imagine you have prepared a qubit in a very specific quantum state. You want another quantum computer, located hundreds of kilometers away, to have **exactly the same quantum state**.

Your first thought might be:

> "Why not just copy the qubit?"

Unfortunately, quantum mechanics doesn't allow this. According to the **No-Cloning Principle**, an unknown quantum state **cannot be copied perfectly**. This creates a major challenge for quantum communication. So how can we send quantum information from one location to another?

The answer is **Quantum Teleportation**.

Instead of sending the physical qubit itself, we transfer **the information contained in the qubit** to another distant qubit.

This protocol is one of the fundamental building blocks of:

- Quantum Networks
- Distributed Quantum Computing
- Quantum Repeaters
- Future Quantum Internet

It demonstrates one of the most fascinating consequences of quantum mechanics: **information can be transferred without physically transporting the original quantum particle.**

Core Concept

What is Quantum Teleportation?

Quantum Teleportation is a protocol that transfers the **quantum state** of one qubit to another distant qubit using:

- a shared entangled pair,
- local quantum operations,
- and two bits of classical communication.

A very important point is that **nothing physical is teleported**.

Only the **state** of the qubit is transferred.

If Alice owns a qubit in the state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ then after teleportation, Bob's qubit becomes

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ even though the original qubit never travels to Bob.

What Actually Gets Teleported?

Many beginners imagine that teleportation moves particles instantly across space. That is **not** what happens. The particle remains where it is. The **quantum information** encoded inside that particle is transferred. Think of it like this. Imagine writing a secret recipe on a piece of paper. Instead of sending the paper itself, you use a special process that destroys the original note while recreating the exact recipe somewhere else. The information survives. The original physical object does not move. Quantum Teleportation works in a similar way.

Why Can't We Simply Copy the Qubit?

Suppose a qubit is in an unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

If we could make a perfect copy, we could create unlimited identical quantum states. This would violate one of the most important principles in quantum mechanics. Therefore, Quantum Teleportation **does not duplicate** a qubit.

Instead, the original quantum state is destroyed during measurement, and the same state appears on Bob's qubit after the protocol is completed. No copies are ever created.

Step-by-Step Understanding

Instead of jumping directly into the circuit, let's first understand the people and qubits involved.

The Three Qubits

The teleportation protocol requires **three qubits**.

Qubit 1 - Unknown State

This is the quantum state Alice wants to send.

$|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$

This state could represent **any valid qubit**. Neither Alice nor Bob needs to know the exact values of α and β.

Qubit 2 - Alice's Entangled Qubit

This qubit forms one half of an entangled pair. Alice keeps this qubit.

Qubit 3 - Bob's Entangled Qubit

Bob owns the second half of the entangled pair. Initially, Bob knows nothing about Alice's unknown state.

The Three Resources Required

Quantum Teleportation always requires three ingredients.

Resource	Purpose
Unknown qubit	The state to be transferred
Shared entangled pair	Quantum communication channel
Classical communication	Sends measurement results to Bob

If even one of these resources is missing, teleportation becomes impossible.

Step 1 - Create an Entangled Pair

Before teleportation begins, Alice and Bob must already share an entangled pair.

This is usually the Bell State

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

This state has a remarkable property. Although the qubits may be far apart, their measurement outcomes remain strongly correlated. The Bell pair acts as the **quantum communication channel**.

Step 2 - Combine the Unknown Qubit with the Bell Pair

Now Alice possesses

- the unknown qubit,
- one entangled qubit,

while Bob possesses

- the other entangled qubit.

Alice performs quantum operations that combine her unknown state with her half of the Bell pair. At this stage, the information contained in the unknown qubit becomes distributed across the three-qubit system. It is no longer stored in a single qubit. This is why teleportation is fundamentally different from simply "sending a particle."

Step 3 - Alice Measures Her Qubits

Alice now measures her two qubits. This is the most surprising part of the protocol. The measurement destroys the original quantum state. After this point, Alice no longer possesses the unknown state. The information has been converted into two classical bits, which tell Bob how to recover the original state. This ensures that no duplicate copy of the state exists, preserving the principles of quantum mechanics.

Visual Flow

The entire Quantum Teleportation protocol can be summarized in the following workflow:

Unknown Quantum State (|ψ⟩)
|
▼
Shared Bell Pair Created

|
▼
Alice Applies CNOT
|
▼
Alice Applies Hadamard
|
▼
Alice Measures Her Two Qubits
|
▼
2 Classical Bits Sent to Bob
|
▼
Bob Applies Required Corrections
|
▼
Bob Recovers the Original State (|ψ⟩)

Essential Mathematics

Before understanding the circuit, let's identify the three quantum states involved.

1. The Unknown Quantum State

Suppose Alice has a qubit in an unknown state: $|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$

where

- **α** and **β** are probability amplitudes.
- They satisfy

$|\alpha|^2+|\beta|^2=1$

This condition ensures that the total probability of measuring the qubit is always **1**.

Remember that Alice does **not** know the values of α and β. If she did, she could simply communicate with them classically. The goal of teleportation is to transfer an **unknown** quantum state.

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2. Shared Bell State

Before teleportation begins, Alice and Bob share the Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

This Bell pair acts as the quantum communication channel between them. The two qubits remain entangled regardless of the distance separating them.

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3. Bob's Final State

After Alice performs her operations and Bob applies the correct quantum gate(s), Bob's qubit becomes $|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$. Notice that this is **identical** to Alice's original state. The original quantum information has been transferred not copied.

Step-by-Step Protocol

Now let's understand exactly what happens during teleportation.

Step 1 - Prepare the Bell Pair

Alice and Bob first create an entangled Bell pair.

This can be done using two familiar gates:

1. Apply a **Hadamard (H)** gate to the first qubit.
2. Apply a **CNOT** gate.

You have already learned both gates individually. Together, they create one of the four Bell states.

Step 2 - Introduce the Unknown State

Alice now has three qubits:

- Qubit 0 → Unknown state
- Qubit 1 → Alice's Bell qubit
- Qubit 2 → Bob's Bell qubit

The unknown qubit has **not** been measured yet.

Step 3 - Entangle the Unknown State

Alice performs:

- CNOT
- Hadamard

These operations mix the unknown state with the Bell pair.

At this stage, the quantum information is spread across the system. It is no longer contained entirely in the first qubit.

Step 4 - Measurement

Alice measures her two qubits.

The result will always be one of four classical outcomes:

Alice Measures	Bob Should Apply
00	Identity (Do Nothing)
01	X Gate
10	Z Gate
11	X then Z

These two bits contain the information Bob needs to reconstruct the original quantum state.

Step 5 - Classical Communication

Alice sends **two classical bits** to Bob. This is an important point. Although entanglement exists instantly, Bob **cannot** recover the quantum state until he receives Alice's classical message.

Therefore, Quantum Teleportation **does not allow faster-than-light communication**. This keeps the protocol fully consistent with Einstein's theory of relativity.

Step 6 - Bob Recovers the State

After receiving Alice's message, Bob performs the required correction gate(s). Immediately after the correction, Bob's qubit becomes $|\psi\rangle$ which is exactly the same quantum state Alice originally wanted to send.

The protocol is now complete.

Worked Example

Suppose Alice wants to send

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

to Bob.

Instead of physically sending the qubit, Alice and Bob already share a Bell pair. Alice performs the teleportation protocol.

Suppose her measurement result is 10

From the correction table, Bob must apply

Z Gate

After applying the Z gate,

Bob's qubit becomes $|+\rangle$

The original state has successfully been transferred. Alice's original state no longer exists because it was destroyed during measurement.

Qiskit Implementation

Step 1 - Import Required Libraries

```
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator
from qiskit import transpile
```

These libraries allow us to create quantum circuits, simulate them, and execute the teleportation protocol.

Step 2 - Create a Three-Qubit Circuit

```
qc = QuantumCircuit(3, 2)
```

- Three quantum qubits
- Two classical bits (for Alice's measurement results)

Step 3 - Prepare the State to Teleport

For demonstration, prepare the |+⟩ state.

```
qc.h(0)
```

Step 4 - Create the Bell Pair

```
qc.h(1)
qc.cx(1,2)
```

Now qubits 1 and 2 are entangled.

Step 5 - Alice's Operations

```
qc.cx(0,1)
qc.h(0)
```

These operations encode the unknown state into the combined three-qubit system.

Step 6 - Measure Alice's Qubits

```
qc.measure(0,0)
qc.measure(1,1)
```

The measurement outcomes determine Bob's correction.

Step 7 - Bob's Conditional Corrections

```
qc.x(2).c_if(qc.cregs[0],1)
qc.z(2).c_if(qc.cregs[0],2)
```

Depending on Alice's measurement results, Bob applies the necessary correction gates.

Complete Program

```
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

qc = QuantumCircuit(3,2)

# Prepare state
qc.h(0)

# Bell pair
qc.h(1)
qc.cx(1,2)

# Teleportation protocol
qc.cx(0,1)
qc.h(0)

# Measurement
qc.measure(0,0)
qc.measure(1,1)

# Conditional corrections
qc.x(2).c_if(qc.cregs[0],1)
qc.z(2).c_if(qc.cregs[0],2)

sim = AerSimulator()
compiled = transpile(qc,sim)

result = sim.run(compiled,shots=1024).result()

print(result.get_counts())
```


Expected Output

When executed, the simulation produces different classical measurement outcomes for Alice's two qubits. Based on these outcomes, Bob applies the corresponding correction gates.

After the correction, Bob's qubit is in the **same quantum state** that Alice originally prepared.

This demonstrates that the **quantum information has been successfully transferred**, even though the original qubit never physically moved from Alice to Bob.

Unlike classical communication, the protocol relies on both **entanglement** and **classical communication**. Without either of these components, teleportation cannot be completed.

Real-World Applications

- **Quantum Networks:** Transfer quantum states between distant quantum processors.
- **Quantum Repeaters:** Extend communication distances by overcoming signal loss.
- **Distributed Quantum Computing:** Enable multiple quantum computers to collaborate on a single computation.
- **Quantum Internet:** Form the foundation for future secure quantum communication networks.
- **Secure Quantum Communication:** Support protocols where quantum information must be transmitted without revealing the underlying state.

Common Beginner Mistakes

❌ **Quantum Teleportation transports particles.**
✔ It transfers the **quantum state**, not the physical particle.

❌ **Teleportation is faster than light.**
✔ Bob must receive Alice's **two classical bits**, so information cannot travel faster than light.

❌ **The original qubit remains unchanged.**
✔ Alice's measurement destroys the original quantum state. No duplicate is created.

❌ **Entanglement alone is enough.**
✔ Both **entanglement** and **classical communication** are required for successful teleportation.

❌ **Quantum Teleportation copies information.**
✔ The protocol transfers the state without violating the **No-Cloning Theorem**.

Connections

Superposition
│
▼
Quantum Gates
│
▼

Multi-Qubit Systems
↓
CNOT Gate
↓
Entanglement
↓
Bell States
↓
Quantum Teleportation

Quantum Teleportation is the first complete protocol where all the concepts learned so far work together to achieve a practical task: transferring an unknown quantum state.

Remember This

- Quantum Teleportation transfers a **quantum state**, not a physical object.
- The protocol requires an **unknown qubit**, a **shared Bell pair**, and **two classical bits**.
- The original quantum state is destroyed during measurement.
- Entanglement provides the quantum resource, while classical communication completes the protocol.
- Teleportation does not violate the speed of light because classical communication is still necessary.
- Bob reconstructs the original state by applying the appropriate correction gates.
- Quantum Teleportation is a foundational protocol for the future **Quantum Internet** and distributed quantum computing.

Math Corner

FORMULA SHEET

1. UNKNOWN QUANTUM STATE

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

where $\alpha, \beta \in \mathbb{C}$ are complex probability amplitudes.
The state is unknown to both Alice and Bob.

2. NORMALIZATION CONDITION

$$|\alpha|^2 + |\beta|^2 = 1$$

The total probability of measuring the qubit in state $|0\rangle$ or $|1\rangle$ must always be 1.

3. BELL STATE (SHARED ENTANGLED PAIR)

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

This is the Bell state shared between Alice and Bob before teleportation begins.

4. COMBINED INITIAL STATE

$$|\Psi_{init}\rangle = |\psi\rangle \otimes |\Phi^+\rangle$$

The 3-qubit system consists of:

- Qubit 0: Unknown state $|\psi\rangle$ (Alice)
- Qubit 1: Alice's half of the Bell pair
- Qubit 2: Bob's half of the Bell pair

5. STATE AFTER ALICE'S OPERATIONS

After applying CNOT $(0 \rightarrow 1)$ and Hadamard (on qubit 0), the state becomes:

$$|\Psi_{after}\rangle = \frac{1}{2}\Big(|00\rangle\,|\psi\rangle + |01\rangle\,X|\psi\rangle + |10\rangle\,Z|\psi\rangle + |11\rangle\,XZ\,|\psi\rangle\Big)$$

where

- X = Pauli-X (bit flip) gate
- Z = Pauli-Z (phase flip) gate
- XZ means first apply Z, then X (or equivalently, $-iY$ up to global phase)

6. POSSIBLE MEASUREMENT OUTCOMES

When Alice measures her two qubits (0 and 1), she gets one of four possible results.
Each result occurs with probability $\frac{1}{4}$.

Alice's Measurement	Bob's Operation on Qubit 2
00	I (Identity) — Do nothing
01	X (Pauli-X gate)
10	Z (Pauli-Z gate)
11	XZ (Pauli-X then Pauli-Z gate)

7. BOB'S RECOVERED STATE

After Bob receives the 2 classical bits from Alice and applies the corresponding operation, his qubit becomes:

$$|\psi_{Bob}\rangle = |\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

Bob's qubit is now in the exact same state as the original unknown state.

8. DIRAC NOTATION REFERENCE

$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix}$$

Computational basis state 0

$$|1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$$

Computational basis state 1

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Equal superposition (plus state)

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Equal superposition (minus state)

Key Takeaway: Quantum Teleportation transfers the state $|\psi\rangle$ from Alice to Bob using entanglement, quantum operations, and 2 bits of classical communication.

End-of-Day Summary

- You learned why quantum states cannot be copied and how Quantum Teleportation overcomes this limitation by transferring the state instead of duplicating it.

- You explored the three essential ingredients of the protocol: an unknown quantum state, a shared Bell pair, and classical communication.
- You understood the sequence of operations performed by Alice and Bob, from Bell pair creation to measurement and state reconstruction.
- You saw how the protocol is implemented in Qiskit and why conditional correction gates are necessary.
- You discovered why Quantum Teleportation is a cornerstone of quantum communication, distributed quantum computing, and the future Quantum Internet.

Top Verified Resources

Official Documentation

- IBM Quantum Learning – Quantum Teleportation: https://quantum.cloud.ibm.com/learning/en/courses/basics-of-quantum-information/entanglement-in-action/quantum-teleportation

Qiskit Tutorial

- Qiskit Community Notebook – Quantum Teleportation: https://github.com/qiskit-community/qiskit-community-tutorials/blob/master/awards/teach_me_quantum_2018/intro2qc/7.Quantum%20teleportation.ipynb

Articles

- TutorialsPoint – Quantum Teleportation in Python: https://www.tutorialspoint.com/article/quantum-teleportation-in-python
- GeeksforGeeks – Quantum Teleportation in Python: https://www.geeksforgeeks.org/python/quantum-teleportation-in-python/

Videos

- Quantum Teleportation Explained: https://www.youtube.com/watch?v=cSubSa3I3f4
- Quantum Teleportation Visual Walkthrough: https://www.youtube.com/watch?v=KsvNsY4cVvE